

Rebuttal

No-Regret Learning in Stackelberg Security Games with an Unknown, Bounded Set of Attacker Types

We thank both reviewers for their careful reading. Below we address every question. Section / equation references are to the submitted report; the labelling issue raised by Reviewer 2 is addressed in R2.Q1/Q3.

Reviewer 1

R1.Q1 (Second Grow-Catalogue variant). We had originally planned a second variant with lazy recomputation of $\mathcal{E}(C_t; \varepsilon)$, but dropped it during implementation. The mentions in Section 7.2 and the corresponding row in Table 1 are leftovers from earlier drafts and will be removed.

R1.Q2 (Validity of the Step 7 bound in our experimental regime). That’s correct. After applying $\log(1+x) \leq x$ in Eq. (17) the correction term we discarded is $K_{\max} n^3 / 2^{n-1}$, which we treated as $O(1)$ on the basis $n^3 / 2^n = O(1)$. This is true asymptotically in n , but is not a numerical bound at small n . The bound should therefore be read as

$$\log N_{\max} = O(n^2 + K_{\max} \log n)$$

holding with an *explicit* multiplicative constant that absorbs the small- n correction. The proof of Theorem 6 uses only the asymptotic form of Eq. (17), so the regret bound is unaffected.

R1.Q3 (Match with Balcan et al. Theorem 5.1). Our bound $\tilde{O}\left(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)}\right)$ does *not* reduce literally to Balcan et al. Theorem 5.1 with $k \leftarrow K_{\max}$. The intended statement is weaker: like their bound, ours is sub-linear in T under an appropriate scaling regime on $K_{\max}$. We make that regime precise in Corollary 7, namely $K_{\max} = o\left(\min(T/n^2, \sqrt{T/\log n})\right)$.

R1.Q4 (Experiments at larger n). We re-ran the per-round-regret experiment at $n = 5$, $K_{\max} = 4$, $T = 6000$ (20 seeds, non-uniform epochs $\ell = (300, 800, 1700, 3200)$), using a sampled ε -net for $\mathcal{E}$ since exact enumeration is too expensive (R1.Q5). The proof of Theorem 6 only requires a finite ε -cover, so the regret bound still holds with a negligible additive εT term. Figure 1 shows the same qualitative picture as $n = 3$: spike at each type-reveal, $O(t^{-1/2})$ decay within each epoch, rolling mean returning to ≈ 0 in the long final epoch.

R1.Q5 (Runtime cost of $\mathcal{E}(C; \varepsilon)$ per discovery round). With $|C| = K$ and n targets, exact enumeration intersects every $(n-1)$ -subset of the $n + K \binom{n}{2}$ defining hyperplanes, giving $T_{\text{exact}}(n, K) = O(K^{n-1} \text{poly}(n))$ for fixed n . The sampling method of R1.Q4 costs only $O(MKn)$ per discovery, so it removes the K^{n-1} blow-up. Measured on a laptop ($n = 5$, mean over 20 seeds):

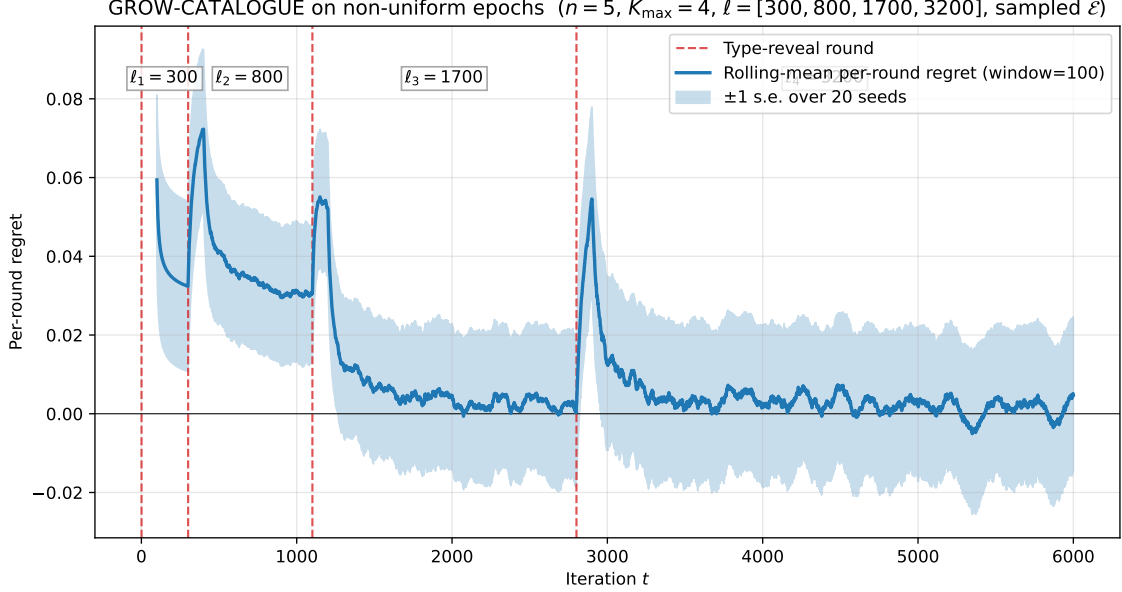


Figure 1: Rebuttal experiment: $n = 5$, $K_{\max} = 4$, $T = 6000$, non-uniform epochs $\ell = (300, 800, 1700, 3200)$, sampled ε -cover ($M = 2000$), mean ± 1 s.e. over 20 seeds.

- Exact enumeration: $K=1$: 80 ms ($|\mathcal{E}|=148$), $K=2$: 6.13 s (1578), $K=3$: 77.7 s (6057), $K=4$: 349 s (13794).
- Sampled ε -net: $K=1$: 4.8 ms ($|\mathcal{E}|\approx 4$), $K=2$: 9.1 ms (10), $K=3$: 13.4 ms (21), $K=4$: 17.6 ms (38).

This cost is paid at most $K_{\max}$ times per run, so amortised over T it is negligible.

R1.Q6 (Definition of q_t). Inside epoch i , let $\mathcal{E}_i = \mathcal{E}(C_i; \varepsilon)$ with $N_i = |\mathcal{E}_i|$, and $L^{(t)}(p) = \sum_{s=1}^{t-1} \ell^{(s)}(p)$ the cumulative loss. The Polynomial-Weights distribution is

$$q_t(p) = \frac{\exp(-\eta_t L^{(t)}(p))}{\sum_{p' \in \mathcal{E}_i} \exp(-\eta_t L^{(t)}(p'))}, \quad \eta_t = \sqrt{\log N_i / t}, \quad (1)$$

the anytime Hedge schedule of [1, 2]. In practice the algorithm keeps the cumulative losses $\{L^{(t)}(p)\}_{p \in \mathcal{E}_i}$ in log space, recomputes η_t each round, normalises via a numerically-stable softmax to obtain q_t , and then draws p_t as a single categorical sample over $\mathcal{E}_i$.

R1.Q7 (Bandit / imperfect-information extension). Extending to bandit feedback (only the attacked target observed, not the type) looks difficult: without type feedback the discovery step is no longer well-defined, so sub-linear regret in full generality seems implausible. A natural direction is to restrict to special classes of attacker types (e.g. well-separated best-response cells or a margin / identifiability condition) under which sub-linear regret may still be achievable.

R1.Q8 (Full publication and main technical gaps). Yes, we are interested in extending this to a full publication. The main directions we would pursue are (i) replacing the exact $\mathcal{E}(C; \varepsilon)$ enumeration with a principled approximate algorithm that is provably sub-linear in regret while

keeping a decent runtime at larger n , and (ii) tightening the dependence on n and $K_{\max}$ in the regret bound.

Reviewer 2

R2.Q1 / R2.Q3 (Numbering of Theorems / Lemmas / Propositions). The labels in the proof of Theorem 6 are inconsistent. The intended cross-references are: “Theorem 1” $\rightarrow$ **Lemma 1**, “Theorem 2” (Step 7) $\rightarrow$ **Lemma 2**, “Theorem 3” (Step 3) $\rightarrow$ **Proposition 3**. “Theorem 1” does not appear as a stated theorem; we will fix the labels in the camera-ready.

R2.Q2 (Number of arms / assumptions in the regret analysis). The number of arms in epoch i is $N_i = |\mathcal{E}(C_i; \varepsilon)|$. We assume (i) bounded per-round losses $\ell^{(t)} \in [0, 1]$ (defender utility is in $[-1, 1]$ by construction), (ii) full-information feedback within an epoch (the attacker’s type is revealed, hence the full loss vector over $\mathcal{E}_i$ is observable), and (iii) Hedge is restarted on the enlarged expert set at every discovery round. The uniform bound $N_i \leq N_{\max}$ from Eq. (17) collapses the per-epoch terms into one $\sqrt{\log N_{\max}}$ factor in Step 10.

R2.Q4 (Notation “PW”). Yes, “PW” is shorthand for the *Polynomial Weights* subroutine, equivalent to Hedge with the anytime learning-rate schedule of (1).

R2 (Additional concern uniform epoch sizes in the original experiments). The reviewer noted that the submitted experiments used uniform epochs. We additionally ran the experiment of Figure 1 with explicitly non-uniform lengths $\ell = (300, 800, 1700, 3200)$. Two points: (i) per-round regret decays like $1/\sqrt{t}$ within an epoch, so short epochs show a higher plateau and long epochs decay cleanly to zero (visible in the figure); (ii) the per-epoch bound depends on ℓ_i only through $\sqrt{\ell_i}$, so by Cauchy-Schwarz the worst case is $K_{\max}$ *equal* epochs non-uniform schedules are always favourable.

References

- [1] N. Cesa-Bianchi, Y. Mansour, G. Stoltz. Improved second-order bounds for prediction with expert advice. *Machine Learning*, 66(2-3):321-352, 2007.
- [2] N. Cesa-Bianchi, G. Lugosi. *Prediction, Learning, and Games*. Cambridge University Press, 2006.